

State-Dependent Uncertainty Bounds for Robust Output-Feedback MPC

Derivation and a Falcon Certificate-Reuse Numerical Example

Research draft

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I Introduction

Robust output-feedback MPC must account for both uncertain state estimates and deviations between an observer trajectory and a nominal prediction. A global uncertainty envelope can be unnecessarily conservative in operating regions with smaller model mismatch. Our objective is to retain an existing offline design while allowing the predicted uncertainty allowance to depend on the operating region covered by the error tubes.

The organization follows [1]: preliminaries, estimation bounds, robust output-feedback MPC, and a numerical example. Joint nominal and tube prediction, a free nominal initial center, and shifted-candidate arguments are established ingredients of Section IV-B of that reference. They are not claimed as new here. Our proposed extension concerns the local process-envelope construction and its incorporation into coupled continuous-time radius dynamics.

We consider robust output-feedback MPC with fixed metrics and observer/controller gains, a fixed regulation target, and time-invariant constraints.

II Preliminaries

A Plant, observer, and nominal prediction

Consider

$$\dot{x} = f_0(x, u) + E_0 D(a(x))w, \quad w \in \mathcal{W}_0, \quad (\text{II.1a})$$

$$y = Cx + F\eta, \quad \eta \in \mathcal{N}, \quad (\text{II.1b})$$

$$\dot{\hat{x}} = f_0(\hat{x}, u) + L(y - C\hat{x}), \quad (\text{II.1c})$$

$$\dot{z} = f_0(z, v), \quad u = v + K(\hat{x} - z). \quad (\text{II.1d})$$

Here v is the nominal control input, not velocity. The function f_0 includes the identified constant bias; only centered residual uncertainty is scaled. Assume the centered disturbance belongs to the componentwise bounded box

$$\mathcal{W}_0 = \{w \in \mathbb{R}^m : |w_j| \leq \bar{w}_j, \ j = 1, \dots, m\}, \quad \bar{w}_j > 0.$$

This set is compact and convex. Its 2^m vertices, denoted $w^{(i)} \in \text{vert}(\mathcal{W}_0)$, are obtained by choosing each component as either $-\bar{w}_j$ or $+\bar{w}_j$. Every point in the box is a convex combination of these vertices. The diagonal matrix $D(a) \in \mathbb{R}^{m \times m}$ scales selected components of the disturbance vector w by $a \in [\beta, 1]$, leaving the remaining components unchanged, with $0 < \beta < 1$. Thus $E_0 D(a) \mathcal{W}_0$ is nested as a increases. The matrix L is a fixed Luenberger observer gain.

Let $P = P_\delta \succ 0$, $X = P^{-1}$, and $T^\top T = P$. Define

$$e = x - \hat{x}, \quad \delta = \hat{x} - z, \quad V_e = e^\top P e, \quad V_e \leq R, \quad \|\delta\|_P \leq s. \quad (\text{II.2})$$

R is a squared radius and $r = \sqrt{R}$ is a radius; neither denotes the true estimation error. The symbol P is distinct from a terminal-cost matrix. Since $x - z = e + \delta$,

$$\|x - z\|_P \leq \sqrt{R} + s. \quad (\text{II.3})$$

Subtracting the dynamics gives

$$\dot{e} = f_0(x, u) - f_0(\hat{x}, u) - LCe + E_0 D(a)w - LF\eta, \quad (\text{II.4a})$$

$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, v) + LCe + LF\eta. \quad (\text{II.4b})$$

Process disturbance enters the estimation error directly. It influences observer-to-nominal tracking indirectly through e ; the observer equation itself contains no additive plant process disturbance.

B Time and prediction notation

In continuous-time expressions such as $x(t)$, $R(t)$, and $s(t)$, the argument t denotes physical time. In the MPC formulation, the subscript t denotes the integer update index; update t occurs at physical time th , where $h > 0$ is the sampling period. In particular, $\hat{x}_t := \hat{x}(th)$, and $\bar{R}_t$ is the estimation-error bound available at that update. Prediction time $\tau \in [0, H_p]$ measures time ahead of the update, with horizon $H_p \geq h$. Thus $z_t(\tau), v_t(\tau), R_t(\tau), s_t(\tau)$ are predictions made at update t for physical time $th + \tau$. Dots denote derivatives with respect to continuous time, or prediction time for predicted trajectories. Time arguments are omitted in general bounds when the meaning is clear. During an applied interval, define $u_t(\tau) := u(th + \tau)$ and $\hat{x}_t(\tau) := \hat{x}(th + \tau)$ for $0 \leq \tau < h$. These are actual interval signals, while $z_t(\tau), v_t(\tau)$ are the planned trajectories; the update estimate satisfies $\hat{x}_t = \hat{x}_t(0)$.

C Domain and incremental conditions

Assumption 1 (Regularity and certificate domain). *The model and observer admit unique solutions for admissible measurable disturbances and controls. The derivative inequalities used below hold throughout a neighborhood of the certified state/input domain $\mathcal{D}$, including every segment needed for incremental arguments. The MPC enforces containment of the true-state tube, observer-state tube, and relevant feedback inputs in this domain at every prediction time.*

For differentiable f_0 , its difference at two states with the same input equals a segment-averaged Jacobian times their difference. Consequently, Jacobian LMIs at isolated grid points do not by themselves establish Assumption 1.

III State-Dependent Estimation Error Bounds

A Recovering the global quadratic certificate

The starting point is the offline robust output-feedback design of Benders et al. [2]. Let $\epsilon \geq 0$ denote the constant estimation-error radius supplied by its global observer certificate, expressed here in the P -norm:

$$\|x - \hat{x}\|_P \leq \epsilon, \quad \text{equivalently} \quad V_e \leq \epsilon^2.$$

This invariant bound holds provided $V_e(0) \leq \epsilon^2$ and the certificate assumptions hold. We use the observer-certificate LMIs implemented in the accompanying ROHMPC repository [3], specialized here to a constant metric and written in the notation of this manuscript. This subsection recovers the original constant radius ϵ and introduces a time-varying squared radius $R(t)$; Section III-B then constructs the local process-envelope reduction. For $b \in \mathbb{R}^n$, introduce the symmetric off-diagonal block

$$\mathcal{S}(b) = \begin{bmatrix} 0 & b \\ b^\top & 0 \end{bmatrix}. \quad (\text{III.1})$$

Let $A_o = A - LC$ and $q_0 = \lambda\epsilon^2$, with $\lambda > 0$. The saved observer certificate has auxiliary matrices W, H satisfying

$$\mathcal{S}(E_0w) \preceq W \quad \forall w \in \mathcal{W}_0, \quad (\text{III.2a})$$

$$\mathcal{S}(-LF\eta) \preceq H \quad \forall \eta \in \mathcal{N}, \quad (\text{III.2b})$$

$$W + H + \begin{bmatrix} XA_o^\top + A_oX + \lambda X & 0 \\ 0 & -q_0 \end{bmatrix} \preceq 0. \quad (\text{III.2c})$$

The last inequality must hold for all relevant Jacobians. Since X is constant, its time derivative is zero. In the implementation W and H are $\bar{W}$ and $\bar{H}$, respectively.

To see explicitly how these LMIs imply the error bound, first consider the global process envelope ($a = 1$). Equation (II.4) can be written as

$$\dot{e} = A_o e + b, \quad b := E_0w - LF\eta.$$

For the nonlinear model, A here is the segment-averaged state Jacobian between $\hat{x}$ and x at the common input u . The following argument applies if the Jacobian LMI holds along that segment. The vector b collects the process disturbance and measurement-noise contributions.

Rewrite the first two inequalities in (III.2) as $\mathcal{S}(E_0w) - W \preceq 0$ and $\mathcal{S}(-LF\eta) - H \preceq 0$, and add the third. The auxiliary matrices cancel:

$$\underbrace{\mathcal{S}(E_0w) - W}_{\preceq 0} + \underbrace{\mathcal{S}(-LF\eta) - H}_{\preceq 0} + \underbrace{W + H + \text{diag}(XA_o^\top + A_oX + \lambda X, -q_0)}_{\preceq 0} \preceq 0.$$

Since $\mathcal{S}$ is linear,

$$\mathcal{S}(E_0w) + \mathcal{S}(-LF\eta) = \mathcal{S}(b),$$

so the resulting matrix inequality is

$$M := \begin{bmatrix} XA_o^\top + A_oX + \lambda X & b \\ b^\top & -q_0 \end{bmatrix} \preceq 0.$$

Negative semidefiniteness means $\xi^\top M \xi \leq 0$ for every vector ξ . Choosing $\xi = [Pe; 1]$ and using $PX = XP = I$ and constant P gives

$$\begin{aligned} \begin{bmatrix} Pe \\ 1 \end{bmatrix}^\top M \begin{bmatrix} Pe \\ 1 \end{bmatrix} &= e^\top P(XA_o^\top + A_oX + \lambda X)Pe + 2e^\top Pb - q_0 \\ &= e^\top (A_o^\top P + PA_o)e + 2e^\top Pb + \lambda e^\top Pe - q_0 \\ &= \dot{V}_e + \lambda V_e - q_0 \leq 0. \end{aligned}$$

Therefore,

$$\dot{V}_e \leq -\lambda V_e + q_0. \quad (\text{III.3})$$

Equivalently, define the invertible matrix $G = \text{diag}(P, 1)$. The congruence transformation of M by G is

$$G^\top M G = \begin{bmatrix} A_o^\top P + PA_o + \lambda P & Pb \\ b^\top P & -q_0 \end{bmatrix} \preceq 0. \quad (\text{III.4})$$

Indeed, $v^\top G^\top M G v = (Gv)^\top M (Gv) \leq 0$ for every v ; invertibility makes the two matrix inequalities equivalent. Evaluating the transformed matrix at $[e; 1]$ is identical to evaluating M at $[Pe; 1]$, because $G[e; 1] = [Pe; 1]$. These inequalities hold for every admissible w, η .

To bound the estimation error, introduce a scalar squared radius $R(t)$ defining the time-varying error set

$$\mathcal{E}(t) = \{e : e^\top P e \leq R(t)\}.$$

Thus $R(t)$ is the upper boundary in the scalar V_e coordinate, and $e^\top P e = R(t)$ is the corresponding boundary in error-state space. Choose $R(0) \geq V_e(0)$ and propagate this boundary using

$$\dot{R} = -\lambda R + q_0.$$

Subtracting this equation from (III.3) gives

$$\frac{d}{dt}(V_e - R) \leq -\lambda(V_e - R), \quad V_e(t) - R(t) \leq e^{-\lambda t}(V_e(0) - R(0)) \leq 0.$$

Consequently, $V_e(t) \leq R(t)$ and $\sqrt{R(t)}$ bounds the estimation error in the P -norm wherever the derivative inequality holds. The comparison equation has solution

$$R(t) = \epsilon^2 + (R(0) - \epsilon^2)e^{-\lambda t}. \quad (\text{III.5})$$

When $R(0) \geq V_e(0)$ it bounds the estimation error; when $R(0) = \epsilon^2$, it exactly recovers the original constant radius ϵ .

B Local process-envelope reduction

Let $W \in \mathbb{S}^{n+1}$ be the process-envelope matrix obtained from the baseline offline SDP in Section III-A, where $\mathbb{S}^{n+1}$ denotes the symmetric $(n+1) \times (n+1)$ matrices. Partition it as

$$W = \begin{bmatrix} Q & h_w \\ h_w^\top & c \end{bmatrix}, \quad Q \succ 0. \quad (\text{III.6})$$

Here $Q \in \mathbb{S}^n$ is the upper-left block, $h_w \in \mathbb{R}^n$ is the off-diagonal block, and $c \in \mathbb{R}$ is the bottom-right entry. These blocks are inherited from the baseline SDP solution. The local construction assumes $Q \succ 0$, keeps Q and h_w fixed, and reduces the scalar entry c to accommodate smaller disturbance sets.

The Schur complement gives

$$\mathcal{S}(E_0 D(a)w) \preceq W \iff (E_0 D(a)w - h_w)^\top Q^{-1} (E_0 D(a)w - h_w) \leq c. \quad (\text{III.7})$$

The matrix Q^{-1} determines the quadratic weighting, h_w the center in the process-disturbance vector space, and c the admissible quadratic level. Define

$$\phi(w, a) := (E_0 D(a)w - h_w)^\top Q^{-1} (E_0 D(a)w - h_w), \quad \phi_i(a) := \phi(w^{(i)}, a).$$

Here ϕ_i is the expression at the fixed vertex $w^{(i)}$, regarded as a function of the scale a . Their pointwise maximum is

$$g(a) = \max_{i=1, \dots, 2^m} \phi_i(a) = \max_{w \in \text{vert}(\mathcal{W}_0)} \phi(w, a). \quad (\text{III.8})$$

For fixed a , the Hessian with respect to w is

$$\nabla_w^2 \phi(w, a) = 2D(a)^\top E_0^\top Q^{-1} E_0 D(a) \succeq 0,$$

so $\phi(\cdot, a)$ is convex. Any $w \in \mathcal{W}_0$ can be written as $w = \sum_i \theta_i w^{(i)}$ with $\theta_i \geq 0$ and $\sum_i \theta_i = 1$. Therefore,

$$\phi(w, a) \leq \sum_i \theta_i \phi(w^{(i)}, a) \leq \max_i \phi_i(a) = g(a).$$

Since the vertices themselves belong to the box, $g(a) = \max_{w \in \mathcal{W}_0} \phi(w, a)$. Thus the vertex maximum bounds the entire disturbance box, including its interior.

For a fixed vertex, the affine dependence of $D(a)$ permits the representation

$$E_0 D(a)w^{(i)} - h_w = ap_i + r_i, \quad \phi_i''(a) = 2p_i^\top Q^{-1} p_i \geq 0,$$

where p_i, r_i are fixed vectors. Each vertex function is consequently convex in a , and their pointwise maximum g is convex as well, although it need not be differentiable where the maximizing vertex changes. Nestedness of $E_0 D(a) \mathcal{W}_0$ also gives $g(\beta) \leq g(1)$.

A convex function lies below the chord joining its endpoint values. Writing $a = \frac{1-a}{1-\beta}\beta + \frac{a-\beta}{1-\beta}1$ for $a \in [\beta, 1]$ yields

$$g(a) \leq \frac{1-a}{1-\beta}g(\beta) + \frac{a-\beta}{1-\beta}g(1).$$

Equivalently,

$$g(a) \leq g(1) - \frac{1-a}{1-\beta}(g(1) - g(\beta)). \quad (\text{III.9})$$

For $\beta < a < 1$, the same inequality can be expressed as a comparison of secant slopes:

$$\frac{g(1) - g(a)}{1-a} \geq \frac{g(1) - g(\beta)}{1-\beta}.$$

The secant ending at 1 and starting at a has slope at least as large as the secant from β to 1. Hence the chord provides a guaranteed reduction from the global value $g(1)$, without requiring derivatives of g . The following proposition transfers this reduction to the process-envelope entry c and the observer forcing q_0 by reducing both by the same amount.

Proposition 1 (Local forcing from a fixed certificate). *Assume (III.2) holds exactly and $Q \succ 0$. Choose $0 \leq \Delta \leq g(1) - g(\beta)$ and define*

$$\theta(a) = \frac{1-a}{1-\beta}, \quad (\text{III.10})$$

$$W(a) = \begin{bmatrix} Q & h_w \\ h_w^\top & c - \Delta\theta(a) \end{bmatrix}, \quad (\text{III.11})$$

$$q(a) = q_0 - \Delta\theta(a). \quad (\text{III.12})$$

Then $W(a)$ covers the local process set and the derivative bound becomes $\dot{V}_e \leq -\lambda V_e + q(a)$.

Proof. At $a = 1$, $D(1) = I$. The process-envelope inequality (III.2a) of the global baseline certificate, together with the Schur-complement equivalence (III.7) and the vertex-maximum identity (III.8), implies

$$c \geq \max_{w \in \mathcal{W}_0} \phi(w, 1) = \max_i \phi_i(1) = g(1).$$

Equation (III.9) and the choice of Δ give $g(a) \leq c - \Delta\theta(a)$, proving the local envelope inequality by (III.7). In the final LMI, reducing the bottom-right entry of W by $\Delta\theta(a)$ is exactly cancelled by replacing $-q_0$ with $-q(a)$. Its other entries are unchanged. The quadratic-form argument for (III.3) then applies. $\square$

At $a = 1$ the global baseline certificate is recovered. This construction changes an auxiliary envelope, not P, K, L , and requires no new SDP. For the subsequent formulation we additionally require $q_0 - \Delta > 0$. For each $a \in [\beta, 1]$, the vertex maximum $g(a)$ covers every process disturbance $E_0 D(a)w$ with $w \in \mathcal{W}_0$. Consequently, the reduced envelope $W(a)$ and the resulting forcing allowance $q(a)$ are valid for all disturbances in the local set $E_0 D(a) \mathcal{W}_0$, with the measurement-noise envelope unchanged.

C Velocity-Dependent Disturbance Bound over the Predicted Tube

Let S_v select the translational velocity components of the state, and denote the true velocity vector by $v_x := S_v x$. The symbol v in the nominal dynamics continues to denote the nominal control input. Consider the velocity-dependent amplitude law

$$a(x) = \beta + (1-\beta) \frac{\|S_v x\|_2^2}{v_{\max}^2}, \quad \|S_v x\|_2 \leq v_{\max}. \quad (\text{III.13})$$

The amplitude enters the physical process disturbance through

$$d(x, w) = E_0 D(a(x))w, \quad w \in \mathcal{W}_0.$$

The diagonal matrix $D(a)$ scales the selected process-disturbance components as defined in Section II-A. Thus a bound on true speed determines an enclosing local process-disturbance set.

Recall that $T^\top T = P$, so $\|e\|_P = \|Te\|_2$. Define

$$c_v := \|S_v T^{-1}\|_2. \quad (\text{III.14})$$

Using the Cauchy–Schwarz inequality and the induced matrix norm bound, we obtain

$$\begin{aligned} \|S_v e\|_2 &= \|S_v T^{-1}(Te)\|_2 \\ &\leq \|S_v T^{-1}\|_2 \|Te\|_2 = c_v \|e\|_P \leq c_v \sqrt{R}. \end{aligned}$$

The same argument gives $\|S_v \delta\|_2 \leq c_v \|\delta\|_P \leq c_v s$. Starting from the state decomposition $x = z + \delta + e$, the triangle inequality gives

$$\begin{aligned} \|S_v x\|_2 &= \|S_v z + S_v \delta + S_v e\|_2 \\ &\leq \|S_v z\|_2 + \|S_v \delta\|_2 + \|S_v e\|_2 \\ &\leq \|S_v z\|_2 + c_v(s + \sqrt{R}). \end{aligned}$$

Combining this tube bound with the certified domain bound $\|S_v x\|_2 \leq v_{\max}$ defines

$$v_{\text{up}} := \min\{v_{\max}, \|S_v z\|_2 + c_v(\sqrt{R} + s)\}. \quad (\text{III.15})$$

Thus v_{up} bounds true speed for every state in the predicted error tube that lies in the certified domain. The cap $v_{\max}$ requires domain containment of the true-state tube; a bound on the nominal speed alone does not establish it.

Since the amplitude law (III.13) is nondecreasing in speed,

$$a(x) \leq a_{\text{up}} := \beta + (1 - \beta)(v_{\text{up}}/v_{\max})^2. \quad (\text{III.16})$$

Nestedness of the scaled disturbance boxes then gives

$$E_0 D(a(x))\mathcal{W}_0 \subseteq E_0 D(a_{\text{up}})\mathcal{W}_0.$$

Applying Proposition 1 to this enclosing set yields

$$\dot{V}_e \leq -\lambda V_e + q(a_{\text{up}}), \quad q(z, R, s) := q(a_{\text{up}}). \quad (\text{III.17})$$

The forcing $q(a_{\text{up}})$ uses an upper bound on the disturbance amplitude over the predicted tube, computed from the nominal velocity and both error radii.

D Comparison with Separate-Norm Disturbance Bounds

Retaining the model, uncertainty sets, metric, and gains introduced in Section II, consider an alternative estimation-error bound obtained by bounding the process-disturbance and measurement-noise contributions separately. Derivative inequalities for error norms are understood to hold almost everywhere in time. If the disturbance-free observer error satisfies $\frac{d}{dt} \|e\|_P \leq -\rho_o \|e\|_P$ for a contraction rate $\rho_o > 0$, a radius bound can be propagated as

$$\dot{r} = -\rho_o r + b_{\text{rest}} + a_{\text{up}} b_{\text{acc}} + b_\eta. \quad (\text{III.18})$$

Here $b_{\text{acc}} = \max \|TE_{\text{acc}}w_{\text{acc}}\|_2$ bounds the unscaled acceleration-disturbance contribution, b_{rest} bounds the remaining process contribution, and $b_\eta = \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2$ bounds measurement noise. Each maximum is taken over its corresponding admissible set.

For vector contributions $d_j(\omega_j)$ with $\omega_j \in \Omega_j$, this construction uses

$$\max_{\omega \in \prod_j \Omega_j} \left\| \sum_j d_j(\omega_j) \right\|_P \leq \sum_j \max_{\omega_j \in \Omega_j} \|d_j(\omega_j)\|_P.$$

The triangle inequality and separate maximizations discard directional information: the individual largest contributions need not align to attain the sum of their norms. Thus this approach can be more conservative and need not recover the original SDP radius at $a = 1$. Proposition 1 retains the original quadratic certificate and recovers its global baseline at $a = 1$.

IV Robust Output-Feedback MPC

A Coupled tube dynamics

Suppose the tracking contraction condition holds along every state-input segment joining (z, v) to $(\hat{x}, v + K(\hat{x} - z))$ in the certified domain:

$$(A + BK)^\top P + P(A + BK) \preceq -2\rho P, \quad \rho > 0. \quad (\text{IV.1})$$

With $T^\top T = P$, define

$$\gamma := \|TLCT^{-1}\|_2, \quad b_\eta := \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2.$$

The constant b_η bounds the measurement-noise contribution $LF\eta$ to the observer dynamics in the P -norm. Equation (II.4) then implies

$$\frac{d}{dt} \|\delta\|_P \leq -\rho \|\delta\|_P + \gamma \sqrt{V_e} + b_\eta. \quad (\text{IV.2})$$

The derivation of (IV.2), using the contraction condition (IV.1) and the Cauchy-Schwarz inequality, is given in Appendix A. Thus the proposed augmented prediction is

$$\dot{z} = f_0(z, v), \quad (\text{IV.3a})$$

$$\dot{R} = -\lambda R + q(z, v, R, s), \quad (\text{IV.3b})$$

$$\dot{s} = -\rho s + \gamma \sqrt{R} + b_\eta. \quad (\text{IV.3c})$$

The general notation permits input dependence in q ; the present velocity example has none explicitly. The forcing bound $q(z, v, R, s)$ must be valid for every state-input pair represented by the predicted tube and every admissible disturbance realization.

Assumption 2 (Radius regularity and initial bounds). *The forcing bound $q(z, v, R, s)$ is locally Lipschitz in its arguments on the set of admissible nominal states, inputs, and radius pairs, and satisfies $0 < q_{\min} \leq q \leq q_{\max}$. Define*

$$0 < R_{\min} \leq q_{\min}/\lambda, \quad R_{\max} = q_{\max}/\lambda, \quad s_{\max} = (\gamma \sqrt{R_{\max}} + b_\eta)/\rho. \quad (\text{IV.4})$$

The pair of radius variables (R, s) is constrained to the rectangle

$$\begin{aligned} \mathcal{B} &= [R_{\min}, R_{\max}] \times [0, s_{\max}] \\ &= \{(R, s) : R_{\min} \leq R \leq R_{\max}, \ 0 \leq s \leq s_{\max}\}. \end{aligned}$$

These variables bound the errors through $e^\top P e \leq R$ and $\|\delta\|_P \leq s$. An available initial prior $\bar{R}_0 \in [R_{\min}, R_{\max}]$ satisfies $V_e(0) \leq \bar{R}_0$. The observer state evolves continuously according to the observer dynamics in Section II, including across MPC updates.

The coupled radius equations (IV.3) satisfy the following conditions on the four boundaries of $\mathcal{B}$:

$$\begin{aligned} R = R_{\min} : \quad \dot{R} &\geq -\lambda R_{\min} + q_{\min} \geq 0, \\ R = R_{\max} : \quad \dot{R} &\leq -\lambda R_{\max} + q_{\max} = 0, \\ s = 0 : \quad \dot{s} &= \gamma\sqrt{R} + b_{\eta} \geq 0, \\ s = s_{\max} : \quad \dot{s} &\leq -\rho s_{\max} + \gamma\sqrt{R_{\max}} + b_{\eta} = 0. \end{aligned}$$

Thus the radius pair cannot leave this rectangle along admissible nominal trajectories: if $(R(0), s(0)) \in \mathcal{B}$, then $(R(t), s(t)) \in \mathcal{B}$ for all subsequent times on which the assumptions hold. Since $\frac{d}{dR}\sqrt{R} = \frac{1}{2\sqrt{R}}$, the derivative is unbounded as $R \rightarrow 0^+$. Restricting $R \geq R_{\min} > 0$ bounds this derivative and ensures that the square-root term is Lipschitz on the admissible interval.

Lemma 1 (Tube containment). *Under Assumptions 1–2, the observer and tracking derivative inequalities, and valid initial bounds, $V_e(t) \leq R(t)$ and $\|\delta(t)\|_P \leq s(t)$ throughout an applied prediction interval.*

Proof. On $V_e = R$, the derivative of $V_e - R$ is nonpositive. Subtracting (IV.3c) from (IV.2) at $\|\delta\|_P = s$, with $V_e \leq R$, gives $\frac{d}{dt}(\|\delta\|_P - s) \leq \gamma(\sqrt{V_e} - \sqrt{R}) \leq 0$. These boundary inequalities, together with the regularity assumptions, ensure that the estimation and tracking errors remain within their coupled bounds. Domain inclusion and a first-exit continuation argument ensure that the derivative hypotheses remain applicable. $\square$

For identical nominal trajectories and initial radii, the local estimation-error forcing bound $q_{\text{local}}(t) \leq q_0$, where $q_0 = \lambda\epsilon^2$ is the global baseline forcing bound, yields $R_{\text{local}}(t) \leq R_{\text{global}}(t)$ and $s_{\text{local}}(t) \leq s_{\text{global}}(t)$. Thus the local formulation produces no larger predicted error bounds under these common conditions. The derivation is given in Appendix B.

B Tightened constraints and MPC initialization

For each state–input constraint $g_j(x, u) \leq 0$, impose the worst-case condition over the predicted error tube:

$$\sup_{\|e\|_P \leq \sqrt{R}, \|\delta\|_P \leq s} g_j(z + \delta + e, v + K\delta) \leq 0. \quad (\text{IV.5})$$

The support function of a set $\mathcal{E}$ is

$$h_{\mathcal{E}}(a) := \sup_{\xi \in \mathcal{E}} a^\top \xi,$$

which gives its largest projection in direction a . For an affine constraint $a_j^\top x + b_j^\top u \leq c_j$, substituting $x = z + \delta + e$ and $u = v + K\delta$ gives the uncertain contribution $(a_j^\top + b_j^\top K)\delta + a_j^\top e$. Evaluating the support functions over the two error balls, with $T^\top T = P$, defines the tightening constants

$$c_{s,j} := \|(a_j^\top + b_j^\top K)T^{-1}\|_2, \quad c_{e,j} := \|a_j^\top T^{-1}\|_2.$$

The norms are Euclidean norms of the indicated row vectors. For fixed affine constraints, metric P , and gain K , these constants can be computed offline. The resulting tightened constraint is

$$a_j^\top z + b_j^\top v + \underbrace{c_{s,j}s}_{\text{tracking-error margin}} + \underbrace{c_{e,j}\sqrt{R}}_{\text{estimation-error margin}} \leq c_j. \quad (\text{IV.6})$$

This expression evaluates the supremum exactly over the chosen product of estimation and tracking error balls. The margins vary along the prediction through R and s . Using the time notation in Section II, the tightened state–input constraints (IV.5) (or (IV.6) for affine

constraints), certificate-domain conditions in Assumption 1, and radius bounds $(R_t, s_t) \in \mathcal{B}$ are imposed for every $\tau \in [0, H_p]$. At update t , minimize the predicted horizon cost

$$J_t = \int_0^{H_p} \ell(z_t(\tau), v_t(\tau)) d\tau + V_f(z_t(H_p)), \quad (\text{IV.7})$$

where ℓ is the stage-cost rate, its integral is the accumulated running cost, and V_f is the terminal cost. Let $\bar{R}_t$ denote the certified squared estimation-error bound available at update t , satisfying $V_e(th) \leq \bar{R}_t$. The prediction satisfies (IV.3), the constraints above, and

$$R_t(0) = \bar{R}_t, \quad \|\hat{x}_t - z_t(0)\|_P \leq s_t(0), \quad (z_t(H_p), R_t(H_p), s_t(H_p)) \in \Omega_f. \quad (\text{IV.8})$$

The initial nominal center $z_t(0)$ is free. For the selected feasible plan, apply

$$u_t(\tau) = v_t(\tau) + K(\hat{x}_t(\tau) - z_t(\tau)), \quad 0 \leq \tau < h,$$

and carry $\bar{R}_{t+1} = R_t(h)$. At the first update, $\bar{R}_0$ is supplied by the initial error bound in Assumption 2. Starting from an initially feasible MPC problem, recursive feasibility and constraint satisfaction hold when the controller applies a feasible plan at each update; optimality is not required for these guarantees. Superscript $*$ denotes optimal plans and the minimum horizon cost in the cost-decrease inequality (IV.10) of Theorem 1.

C Terminal Ingredients and Recursive Feasibility

Assumption 3 (Terminal ingredients). *There exist a nonempty nominal terminal set Z_f and controller κ_f such that $\dot{z} = f_0(z, \kappa_f(z))$ leaves Z_f invariant. For every $z \in Z_f$, with nominal input $v = \kappa_f(z)$, the tightened state-input constraints (IV.5) and the certificate-domain conditions in Assumption 1 hold at $R = R_{\max}$ and $s = s_{\max}$. For the cost assertion additionally assume $\ell, V_f \geq 0$ and*

$$\nabla V_f(z) f_0(z, \kappa_f(z)) \leq -\ell(z, \kappa_f(z)). \quad (\text{IV.9})$$

Then $\Omega_f = Z_f \times \mathcal{B}$ is invariant under terminal control and the radius dynamics. The terminal set is certified using the largest admissible estimation and tracking radii. Since the smaller error balls are contained in these largest ones, this ensures terminal constraint satisfaction and certificate-domain containment for every $(R, s) \in \mathcal{B}$, while the predictions along the horizon retain their local, time-varying radii.

Theorem 1 (Recursive Feasibility and Robust Constraint Satisfaction). *Suppose Assumptions 1–3 and the estimation-error inequality (III.17), written as $\dot{V}_e \leq -\lambda V_e + q(z, v, R, s)$, and tracking-error inequality (IV.2) hold exactly, and the initial MPC problem is feasible. Applying a feasible plan satisfying (IV.8) and carrying $\bar{R}_{t+1} = R_t(h)$ at each update gives recursive feasibility and continuous-time robust constraint satisfaction for all disturbances in the assumed sets. If (IV.9) holds and the selected feasible plans attain the minimum horizon cost among all feasible plans, then*

$$J_{t+1}^* \leq J_t^* - \int_0^h \ell(z_t^*(\tau), v_t^*(\tau)) d\tau. \quad (\text{IV.10})$$

Proof. Feasible candidate at the next update. Suppose the plan (z_t, v_t, R_t, s_t) is feasible at update t . For $0 \leq \tau \leq H_p - h$, define

$$(z^c, v^c, R^c, s^c)(\tau) = (z_t, v_t, R_t, s_t)(\tau + h).$$

The prescribed bound update and Lemma 1 give

$$R^c(0) = R_t(h) = \bar{R}_{t+1}, \quad \|\hat{x}_{t+1} - z^c(0)\|_P = \|\hat{x}_{t+1} - z_t(h)\|_P \leq s_t(h) = s^c(0).$$

Thus both initialization conditions in (IV.8) hold. On this shifted interval the candidate reproduces the previous feasible trajectory, so the time-invariant dynamics (IV.3), tightened constraints (IV.5), certificate-domain conditions, and radius bounds remain satisfied.

At $\tau = H_p - h$, the candidate reaches the previous terminal point in Ω_f . Over the remaining h units of time, apply $v^c = \kappa_f(z^c)$ and continue the nominal and radius dynamics. Terminal invariance under Assumption 3 and invariance of $\mathcal{B}$ keep this continuation in Ω_f , where all tightened and domain constraints hold. Its endpoint therefore satisfies the new terminal condition. This constructs a feasible plan at update $t + 1$; induction from initial feasibility proves recursive feasibility.

Actual constraint satisfaction. During each applied interval, Lemma 1 gives $\|e\|_P \leq \sqrt{R_t(\tau)}$ and $\|\delta\|_P \leq s_t(\tau)$. The actual pair satisfies $x = z_t + \delta + e$ and $u = v_t + K\delta$, and hence belongs to the set over which (IV.5) takes its supremum. Consequently,

$$g_j(x, u) \leq \sup_{\|e\|_P \leq \sqrt{R_t(\tau)}, \|\delta\|_P \leq s_t(\tau)} g_j(z_t(\tau) + \delta + e, v_t(\tau) + K\delta) \leq 0.$$

This holds throughout every applied interval and for all admissible disturbances.

Optimal-cost decrease. Now take an optimal plan at update t . Let $z_f(\sigma)$, $0 \leq \sigma \leq h$, be the appended terminal trajectory, with $z_f(0) = z_t^*(H_p)$ and input $\kappa_f(z_f)$. The shifted candidate has cost

$$\begin{aligned} J_{t+1}^c &= \int_h^{H_p} \ell(z_t^*(\tau), v_t^*(\tau)) d\tau \\ &\quad + \int_0^h \ell(z_f(\sigma), \kappa_f(z_f(\sigma))) d\sigma + V_f(z_f(h)). \end{aligned}$$

Integrating (IV.9) along the terminal continuation gives

$$\int_0^h \ell(z_f(\sigma), \kappa_f(z_f(\sigma))) d\sigma + V_f(z_f(h)) \leq V_f(z_t^*(H_p)).$$

The terminal-cost decrease therefore covers the accumulated stage cost of the appended interval. Substituting this bound and comparing with (IV.7) yields

$$\begin{aligned} J_{t+1}^c &\leq \int_h^{H_p} \ell(z_t^*(\tau), v_t^*(\tau)) d\tau + V_f(z_t^*(H_p)) \\ &= J_t^* - \int_0^h \ell(z_t^*(\tau), v_t^*(\tau)) d\tau. \end{aligned}$$

Since the next optimal cost is no greater than the cost of this feasible candidate, $J_{t+1}^* \leq J_{t+1}^c$, which establishes (IV.10). $\square$

The shifted-and-extended candidate provides a feasible fallback. If a suboptimal solver returns a feasible plan whose horizon cost does not exceed the candidate cost, the same argument establishes the cost-decrease inequality for the selected plans.

Remark 1 (Relation to Section IV of [1]). Section IV-B of [1], Eq. (26), jointly predicts the nominal state/input and estimation and tracking bounds with a free initial nominal center. Using z for that reference's nominal state, its Eq. (26b) and the candidate construction in Theorem 4 of [1] give

$$z_{0|t+1}^c = z_{1|t}^*, \quad \bar{s}_{0|t+1}^c = V_\delta(z_{1|t}^*, \hat{x}_{t+1}) \leq \bar{s}_{1|t}^*.$$

The online estimation update also ensures $\bar{e}_{t+1} \leq \bar{e}_{1|t}^*$. Monotonic propagation of these bounds preserves inclusion in the previously predicted tubes. Here $k|t$ denotes prediction step k at update t in the discrete-time notation of [1]. Its bounds $\bar{e}, \bar{s}$ use general Lyapunov functions and need not share the numerical scaling of our R, s .

Our continuous-time formulation carries the estimation bound and permits the shifted candidate

$$\bar{R}_{t+1} = R_t(h), \quad z_{t+1}^c(0) = z_t(h), \quad s_{t+1}^c(0) = s_t(h).$$

This construction does not exploit additional reductions in the available estimation bound or the measured initial tracking mismatch. It simplifies the feasibility proof but can retain larger initial bounds than a scheme that incorporates those reductions. The optimizer remains free to choose a smaller feasible tracking radius. This comparison concerns the bound-update mechanism; the proposed state-dependent forcing can reduce future bounds.

Section IV-C of [1], Eq. (32b), instead resets the nominal center:

$$z_{0|t} = \hat{x}_t, \quad \bar{s}_{0|t} = 0.$$

Theorem 5 of [1] establishes recursive feasibility for that formulation using a modified candidate and the terminal conditions in Assumption 9 of [1]. The present theorem treats the free-center formulation.

The horizon-cost decrease in (IV.10) bounds the accumulated nominal running cost. By (II.3), the true-state deviation satisfies

$$\|x - x_{\text{ref}}\|_P \leq \|z - x_{\text{ref}}\|_P + \sqrt{R} + s.$$

Consequently, if the applied nominal trajectory converges to x_{ref} , then

$$\limsup_{t \rightarrow \infty} \|x(t) - x_{\text{ref}}\|_P \leq \sqrt{R_{\text{max}}} + s_{\text{max}},$$

where t denotes continuous time. Nominal convergence is an additional condition here; it is not established by the cost-decrease inequality alone.

V Numerical Example: Falcon Model

A Model and reproduced offline design

The example uses the repository [3] and the saved Falcon design from 10 September 2026. The 12-state model consists of position, attitude, translational velocity, and body angular velocity, with four rotor inputs. The nominal translational drag matrix is zero in this instance. Its Jacobians are therefore independent of translational velocity; this does not restrict the controller to hovering. The velocity domain is $[-2, 2]^3$ m/s, giving $v_{\text{max}} = \sqrt{12}$ m/s.

The original 1024-point Jacobian grid varies four rotor inputs, three angles, and three body rates at two endpoints each. Positions and translational velocities are zero at these points. This grid alone does not certify the continuous domain of the remaining nonlinear coordinates.

The reproduced robust-design and terminal-cost SDPs returned **OPTIMAL**, with objectives 1494.858118 and 38.564610, respectively. The first is a weighted combination of normalized squared tightening coefficients and squared estimation radius, not an optimized contraction rate. The second minimizes the trace of the terminal-cost matrix. Solver success does not ensure useful tightened constraints: the reproduced motor-input half-width was approximately 0.1200, whereas the associated global margin was 0.4619, leaving negative available width. Consequently, this design has not supplied the nonempty terminal set required by Theorem 1.

B Prescribed local uncertainty and parameters

For the Falcon state ordering (position, attitude, translational velocity, body angular velocity), the velocity selector and disturbance scaling matrix are

$$S_v = \begin{bmatrix} 0_{3 \times 6} & I_3 & 0_{3 \times 3} \end{bmatrix}, \quad D(a) = \text{diag}(I_6, aI_3, I_3).$$

The scaled disturbance components enter the translational acceleration equations. Only the centered acceleration disturbance components 7–9 are scaled. Other process-disturbance components, measurement noise, and the nominal bias remain unchanged. We use $\beta = 0.25$ and the velocity law in Section III-C. This is a synthetic uncertainty model, not a new identification or UQ result. The experiment reuses the saved $X, P, L, K, W, H, \lambda$ and solves no new SDP. The factor 0.99

Table 1: Saved certificate and local-envelope parameters (rounded).

Quantity	Value
Observer decay coefficient λ	0.497029
Original uncalibrated ϵ	0.07442638619
Global forcing q_0	0.002753187883
$g(1)$	0.00148773
$g(\beta)$	0.00104178
$\Delta = 0.99[g(1) - g(\beta)]$	0.000441487
$q(\beta)$	0.0023117
Condition number of Q	3474.67

retains a numerical cushion but does not repair an infeasible baseline certificate or provide a rigorous floating-point error bound.

C Radius propagation experiment

A prescribed nominal speed is 0 m/s on $[0, 8)$ s, 2.5 m/s on $[8, 16)$ s, and 0 m/s on $[16, 32]$ s. Initialize $R(0) = \epsilon^2$, $s(0) = 0$. Both cases use identical metrics, gains, initial bounds, tracking dynamics, and noise; only the local forcing versus q_0 changes. The global estimation radius equals ϵ by (III.5). The final estimation-radius reduction is $100(1 - r_{\text{local}}/\epsilon) = 6.68019\%$. This is

Table 2: Final-time comparison at 32 s; these are bounds, not measured errors.

Quantity	Global	State-dependent
Estimation radius r	0.07442638619	0.06945456129
Tracking radius s	0.392384	0.367532

neither a time-average improvement nor a universal performance guarantee. The speed schedule is not a dynamically feasible quadrotor trajectory, and no MPC optimization or closed-loop plant simulation is performed. The earlier separate-norm prototype produced a global equilibrium radius of approximately 0.105718 and a final local radius of 0.0979251, both above the original 0.0744264. These values belong to a different bound construction and do not indicate empirical calibration of the original ϵ . The quadratic-envelope approach restores a baseline-equivalent comparison.

D Numerical checks and limitations

A numerical MPC implementation that checks constraints only at selected prediction times must additionally establish constraint satisfaction between those times. The continuous-time formulation in Section IV-B imposes them throughout the horizon. The current experiment checks all 4096 original process vertices, all noise vertices, the 1024 original dynamics points, and all process vertices at 31 scale values. It additionally evaluates 1968 admissible random nonlinear error configurations. Table 3 reports the saved residuals. The saved feasibility tolerance is 10^{-6} . Small positive residuals mean feasibility within that numerical tolerance, not exact semidefinite feasibility. The convex-chord argument covers intermediate scales analytically only conditional

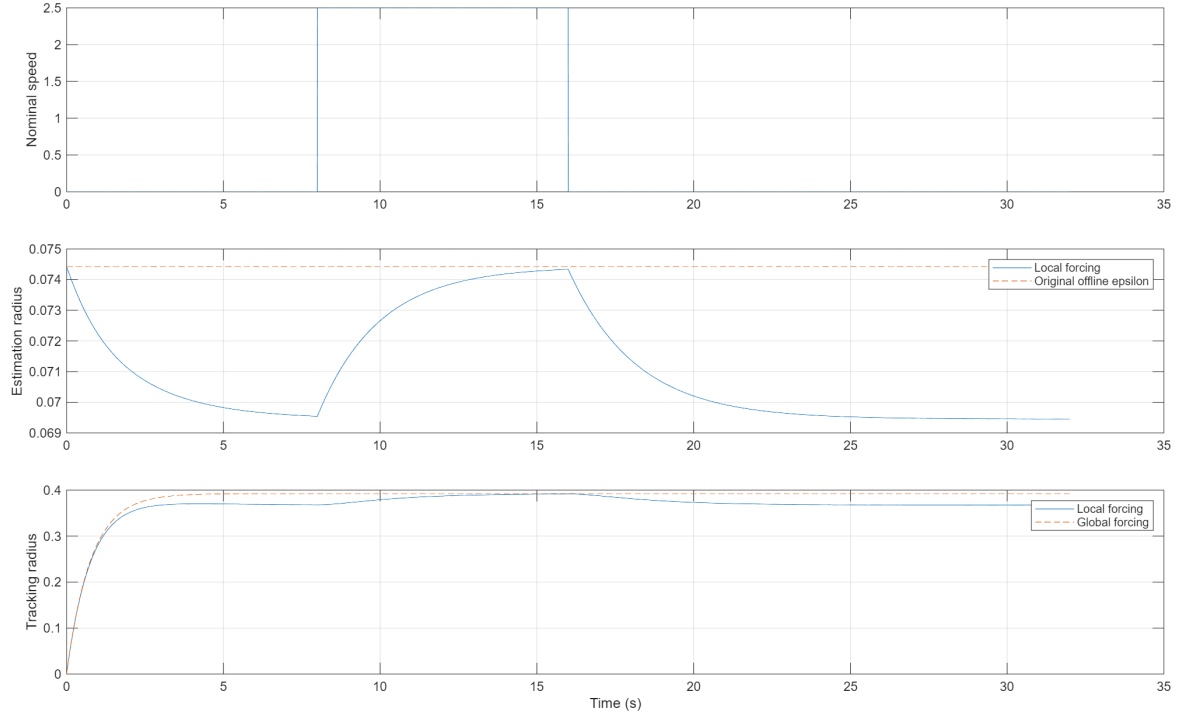


Figure 1: Recorded tube-ODE experiment with a prescribed low/high/low nominal-speed schedule. The figure is copied from the saved run, not generated by a closed-loop simulator.

Table 3: Numerical audit; positive matrix residuals are tolerated numerically, not exact certificates.

Check	Maximum residual
Original process-envelope LMI	$+5.50832 \cdot 10^{-8}$
Original noise-envelope LMI	$+5.57035 \cdot 10^{-8}$
Observer dynamics LMI	$+5.45468 \cdot 10^{-8}$
Tracking LMI	-0.0030671
Local Schur envelope	$+5.63813 \cdot 10^{-8}$
Sampled nonlinear derivative excess	-0.00229595

on exact baseline inequalities. Finite Jacobian and nonlinear samples do not prove full-domain coverage. No closed-loop feasibility, safety, or stability claim for Falcon follows from these tests.

VI Conclusion

A fixed observer certificate can support locally reduced process forcing through a Schur-complement envelope construction. Coupling this forcing to a tracking radius yields a candidate state-dependent output-feedback MPC formulation with conditional recursive-feasibility and safety guarantees. The Falcon radius experiment demonstrates a smaller local bound relative to a baseline-equivalent global propagation. Remaining work is to certify the continuous domain and numerical margins, validate the local uncertainty model, construct compatible nonempty terminal ingredients, and then implement and evaluate the closed-loop controller.

References

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- [3] D. Benders et al., ROHMPC code repository, <https://github.com/dbenders1/rohmpc>. Numerical inputs and saved outputs from the local reproduction of 10 September 2026.

APPENDIX

A Derivation of the Tracking-Error Bound

Recall $\delta = \hat{x} - z$, $e = x - \hat{x}$, and $u = v + K\delta$. Subtracting the nominal dynamics from the observer dynamics gives

$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, v) + LCe + LF\eta. \quad (\text{A.1})$$

For fixed z, v, δ , parameterize the straight-line state-input segment by

$$x_\theta = z + \theta\delta, \quad u_\theta = v + \theta K\delta, \quad 0 \leq \theta \leq 1.$$

Here θ parameterizes a segment in state-input space, not time. Its endpoints are (z, v) and $(\hat{x}, u)$. With $A = \partial f_0 / \partial x$ and $B = \partial f_0 / \partial u$, integration along this segment yields

$$f_0(\hat{x}, u) - f_0(z, v) = \int_0^1 [A(x_\theta, u_\theta) + B(x_\theta, u_\theta)K] \delta d\theta \quad (\text{A.2})$$

$$= \bar{A}_K \delta, \quad \bar{A}_K := \int_0^1 [A(x_\theta, u_\theta) + B(x_\theta, u_\theta)K] d\theta. \quad (\text{A.3})$$

The domain assumptions ensure the contraction inequality holds at every point of this segment. Integrating it and using constant P gives

$$\bar{A}_K^\top P + P \bar{A}_K \preceq -2\rho P. \quad (\text{A.4})$$

Define $d_o := LCe + LF\eta$. Then (A.1) becomes $\dot{\delta} = \bar{A}_K \delta + d_o$. For $\ell_\delta := \|\delta\|_P = \sqrt{\delta^\top P \delta} > 0$,

$$\dot{\ell}_\delta = \frac{\delta^\top P \dot{\delta}}{\ell_\delta} \quad (\text{A.5})$$

$$= \frac{\delta^\top P \bar{A}_K \delta}{\ell_\delta} + \frac{\delta^\top P d_o}{\ell_\delta}. \quad (\text{A.6})$$

The contraction contribution satisfies

$$\delta^\top P \bar{A}_K \delta = \frac{1}{2} \delta^\top (P \bar{A}_K + \bar{A}_K^\top P) \delta \leq -\rho \ell_\delta^2.$$

For the forcing contribution, $T^\top T = P$ and the Cauchy-Schwarz inequality imply

$$\frac{\delta^\top P d_o}{\ell_\delta} = \frac{(T\delta)^\top (Td_o)}{\|T\delta\|_2} \leq \|Td_o\|_2.$$

The triangle inequality and the definitions of γ and b_η give

$$\|Td_o\|_2 \leq \|TLCe\|_2 + \|TLF\eta\|_2 \quad (\text{A.7})$$

$$\leq \|TLC T^{-1}\|_2 \|Te\|_2 + \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2 \quad (\text{A.8})$$

$$= \gamma \sqrt{V_e} + b_\eta. \quad (\text{A.9})$$

Combining these estimates establishes

$$\frac{d}{dt} \|\delta\|_P \leq -\rho \|\delta\|_P + \gamma \sqrt{V_e} + b_\eta, \quad (\text{A.10})$$

which is (IV.2). The calculation involving division by ℓ_δ applies when $\delta \neq 0$. Along absolutely continuous trajectories, the norm is absolutely continuous and its derivative is zero almost

everywhere on the set where the norm vanishes. Since the forcing bound is nonnegative, the same inequality therefore holds almost everywhere in time, including that zero set.

Finally, $V_e \leq R$ gives the comparison dynamics

$$\dot{s} = -\rho s + \gamma\sqrt{R} + b_\eta, \quad s(0) \geq \|\delta(0)\|_P.$$

Subtracting this equation from the norm inequality and applying an integrating factor yields

$$\|\delta(t)\|_P - s(t) \leq e^{-\rho t}(\|\delta(0)\|_P - s(0)) \leq 0,$$

provided the estimation bound and certificate-domain conditions remain valid, as required in the coupled containment argument.

B Comparison of Local and Global Radius Bounds

Consider the same nominal state and input trajectories $z(t), v(t)$, identical initial radius values, and the same metric, gains, and constants $\lambda, \rho, \gamma, b_\eta$ in both propagations. Define the local estimation-error forcing along its radius trajectory by

$$q_{\text{local}}(t) := q(z(t), v(t), R_{\text{local}}(t), s_{\text{local}}(t)).$$

For the envelope construction in Proposition 1, $q_{\text{local}}(t) \leq q_0$, where $q_0 = \lambda\epsilon^2$ is the global baseline forcing bound. The comparison below holds on any interval on which both propagations are well-defined and their domain assumptions hold.

The squared estimation radii satisfy

$$\dot{R}_{\text{local}} = -\lambda R_{\text{local}} + q_{\text{local}}(t), \tag{B.1}$$

$$\dot{R}_{\text{global}} = -\lambda R_{\text{global}} + q_0, \quad R_{\text{local}}(0) = R_{\text{global}}(0). \tag{B.2}$$

Let $\Delta_R = R_{\text{local}} - R_{\text{global}}$. Subtracting the equations and multiplying by $e^{\lambda t}$ gives

$$\frac{d}{dt}(e^{\lambda t}\Delta_R(t)) = e^{\lambda t}(q_{\text{local}}(t) - q_0).$$

Integration, using $\Delta_R(0) = 0$, yields

$$R_{\text{local}}(t) - R_{\text{global}}(t) = \int_0^t e^{-\lambda(t-\tau)}(q_{\text{local}}(\tau) - q_0) d\tau \leq 0. \tag{B.3}$$

The inequality follows because the exponential is positive and the forcing difference is nonpositive. This argument permits q_{local} to depend on the local radius trajectory; it only uses its upper bound by q_0 .

The corresponding tracking radii satisfy

$$\dot{s}_{\text{local}} = -\rho s_{\text{local}} + \gamma\sqrt{R_{\text{local}}} + b_\eta, \tag{B.4}$$

$$\dot{s}_{\text{global}} = -\rho s_{\text{global}} + \gamma\sqrt{R_{\text{global}}} + b_\eta, \quad s_{\text{local}}(0) = s_{\text{global}}(0). \tag{B.5}$$

Subtracting and applying the integrating factor $e^{\rho t}$ similarly gives

$$\begin{aligned} s_{\text{local}}(t) - s_{\text{global}}(t) &= \gamma \int_0^t e^{-\rho(t-\tau)} \left(\sqrt{R_{\text{local}}(\tau)} - \sqrt{R_{\text{global}}(\tau)} \right) d\tau \\ &\leq 0. \end{aligned} \tag{B.6}$$

Here (B.3), nonnegative radii, and $\gamma \geq 0$ establish the final inequality. Consequently, under the common nominal trajectories and initialization, both predicted error bounds in the local formulation are no larger than their global baseline counterparts.